

3) To perform data manipulations (INSERT, DELETE, UPDATE) and use aggregation functions (SUM, COUNT, MAX) on the DEPOSIT, BORROW, EMPLOYEE, and JOB tables, with sorting operations.

- Create table
- Deposit table

Create table deposit(account_no number primary key, cust_name varchar2(20), balance number, branch varchar2(20));

- Insert into deposit values(101, 'ravi shah', 25000, 'maninagar');
- Insert into deposit values(102, 'abc', 18000, 'navrangpura');
- Insert into deposit values(103, 'xyz', 40000, 'bapunagar');
- Insert into deposit values(104, 'pqr', 15000, 'maninagar');

```
SQL> create table deposit(account_no number primary key, cust_name varchar2(20), balance number, branch varchar2(20));
Table created.
SQL> insert into deposit values(101, 'ravi shah', 25000, 'maninagar');
1 row created.
SQL> insert into deposit values(102, 'abc', 18000, 'navrangpura');
1 row created.
SQL> insert into deposit values(103, 'xyz', 40000, 'bapunagar');
1 row created.
SQL> insert into deposit values(104, 'pqr', 15000, 'gota');
1 row created.
```

- Borrow table

Create table borrow(loan_no number primary key, cust_name varchar2(20), amount number, branch varchar2(20));

- Insert into borrow values(201, 'ravi shah', 20000, 'maninagar');
- Insert into borrow values(202, 'abc', 8000, 'navrangpura');
- Insert into borrow values(203, 'xyz', 30000, 'bapunagar');
- Insert into borrow values(204, 'pqr', 12000, 'maninagar');

```

SQL> create table borrow(loan_no number primary key,cust_name varchar2(20),amount number,branch varchar2(20));
Table created.
SQL> insert into borrow values(201,'ravi shah',20000,'maninagar');
1 row created.
SQL> insert into borrow values(202,'abc',8000,'navrangpura');
1 row created.
SQL> insert into borrow values(203,'xyz',30000,'bapunagar');
1 row created.
SQL> insert into borrow values(204,'pqr',12000,'gota');
1 row created.

```

- Employee table

Create table employee(emp_id number primary key,emp_name varchar2(20),job_id number,salary number,hire_date date,dept_no number);

- Insert into employee values(110,'ravi shah',1,30000,to_date('2020-01-10','yyyy-mm-dd'),20);
- Insert into employee values(111,'abc',2,35000,to_date('2029-07-15','yyyy-mm-dd'),20);
- Insert into employee values(112,'xyz',3,28000,to_date('2021-03-20','yyyy-mm-dd'),20);
- Insert into employee values(113,'pqr',4,40000,to_date('2018-09-12','yyyy-mm-dd'),20);

```

SQL> create table employees(emp_id number primary key,emp_name varchar2(20),job_id number,salary number(10,2),hire_date date,dept_no number);
Table created.
SQL> insert into employees values(110,'ravi shah',1,30000,to_date('2020-01-10','yyyy-mm-dd'),20);
1 row created.
SQL> insert into employees values(111,'abc',2,35000,to_date('2029-07-15','yyyy-mm-dd'),20);
1 row created.
SQL> insert into employees values(112,'xyz',3,28000,to_date('2021-03-20','yyyy-mm-dd'),20);
1 row created.
SQL> insert into employees values(113,'pqr',4,40000,to_date('2018-09-12','yyyy-mm-dd'),20);
1 row created.

```

- Job table

Create table job(job_id number primary key,job_title varchar2(20),min_salary number(10,2),max_salary number(10,2));

- Insert into job values(1,'clerk',10000,30000);
- Insert into job values(2,'manager',20000,60000);
- Insert into job values(3,'analyst',25000,50000);
- Insert into job values(4,'officer',30000,70000);

```
SQL> create table jobs(job_id number primary key,job_title varchar2(20),min_salary number(10,2),max_salary number(10,2));
Table created.
SQL> insert into jobs values(1,'clerk',10000,30000);
1 row created.
SQL> insert into jobs values(2,'manager',20000,60000);
1 row created.
SQL> insert into jobs values(3,'analyst',25000,50000);
1 row created.
SQL> insert into jobs values(4,'officer',30000,70000);
1 row created.
```

- Table before update
- Deposit table

```
SQL> select * from deposit;
```

ACCOUNT_NO	CUST_NAME	BALANCE	BRANCH
101	ravi shah	25000	maninagar
102	abc	18000	navrangpura
103	xyz	40000	bapunagar
104	pqr	15000	gota

- Borrow table

```
SQL> select * from borrow;
```

LOAN_NO	CUST_NAME	AMOUNT	BRANCH
201	ravi shah	20000	maninagar
202	abc	8000	navrangpura
203	xyz	30000	bapunagar
204	pqr	12000	gota

- Employee table

```
SQL> select * from employees;
```

EMP_ID	EMP_NAME	JOB_ID	SALARY	HIRE_DATE	DEPT_NO
110	ravi shah	1	30000	10-JAN-20	20
111	abc	2	35000	15-JUL-29	20
112	xyz	3	28000	20-MAR-21	20
113	pqr	4	40000	12-SEP-18	20

- Job table

```
SQL> select * from jobs;
```

JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
1	clerk	10000	30000
2	manager	20000	60000
3	analyst	25000	50000
4	officer	30000	70000

- Questions

1) Increase the salary of all employees in Dept_No = 20 by 7%.

```
SQL> update employees
  2 set salary = salary * 1.07
  3 where dept_no = 20;
```

4 rows updated.

```
SQL> select * from employees;
```

EMP_ID	EMP_NAME	JOB_ID	SALARY	HIRE_DATE	DEPT_NO
110	ravi shah	1	32100	10-JAN-20	20
111	abc	2	37450	15-JUL-29	20
112	xyz	3	29960	20-MAR-21	20
113	pqr	4	42800	12-SEP-18	20

2) Update the balance of customer 'bhavya' to 35,000.

```
SQL> update deposit
  2 set balance = 35000
  3 where cust_name = 'ravi shah';
```

1 row updated.

```
SQL> select * from deposit;
```

ACCOUNT_NO	CUST_NAME	BALANCE	BRANCH
101	ravi shah	35000	maninagar
102	abc	18000	navrangpura
103	xyz	40000	bapunagar
104	pqr	15000	gota

3) Reduce the loan amount of all customers from branch 'Maninagar' by 5%.

```
SQL> update borrow
  2 set amount = amount * 0.95
  3 where branch = 'maninagar';
```

1 row updated.

```
SQL> select * from borrow;
```

LOAN_NO	CUST_NAME	AMOUNT	BRANCH
201	ravi shah	19000	maninagar
202	abc	8000	navrangpura
203	xyz	30000	bapunagar
204	pqr	12000	gota

4) Delete the employee whose Emp_ID = 110.

```
SQL> delete from employees
2 where emp_id = 110;
```

1 row deleted.

```
SQL> select * from employees;
```

EMP_ID	EMP_NAME	JOB_ID	SALARY	HIRE_DATE	DEPT_NO
111	abc	2	37450	15-JUL-29	20
112	xyz	3	29960	20-MAR-21	20
113	pqr	4	42800	12-SEP-18	20

5) Delete all depositors from branch 'Navrangpura'.

```
SQL> delete from deposit
2 where branch = 'navrangpura';
```

1 row deleted.

```
SQL> select * from deposit;
```

ACCOUNT_NO	CUST_NAME	BALANCE	BRANCH
101	ravi shah	35000	maninagar
103	xyz	40000	bapunagar
104	pqr	15000	gota

6) Delete loan records where the amount is less than 10,000.

```
SQL> delete from borrow
2 where amount < 10000;
```

1 row deleted.

```
SQL> select * from borrow;
```

LOAN_NO	CUST_NAME	AMOUNT	BRANCH
201	ravi shah	19000	maninagar
203	xyz	30000	bapunagar
204	pqr	12000	gota

7) Find the total deposit amount across all branches.

```
SQL> select sum(balance) as total_deposit
2   from deposit;

TOTAL_DEPOSIT
-----
          90000
```

8) Find the highest salary in the EMPLOYEE table.

```
SQL> select max(salary) as highest_salary
2   from employees;

HIGHEST_SALARY
-----
          42800
```

9) Count how many employees are working under each Job_ID.

```
SQL> select job_id, count(*) as employees_count
2   from employees
3  group by job_id;

JOB_ID EMPLOYEES_COUNT
-----
      2                1
      3                1
      4                1
```

10) Find the average loan amount of each branch from the BORROW table.

```
SQL> select branch, avg(amount) as avg_loan
2   from borrow
3  group by branch;

BRANCH                AVG_LOAN
-----
maninagar              19000
bapunagar              30000
gota                   12000
```

